Comparative Assessment: WXIMPACTBENCH vs. CRISISBENCH

Deployment context: Argentine Government Disaster Reporting Reliability — Policy Staff

Deployment Context

Use case: Use case: Reliability of disaster reporting in news at national/local level, finding alignment with effective and verified information (e.g. noisy vs real signals). The evaluation data could report or not disaster damage.

Target population: Policy makers and government agencies in Argentina studying trustiness and reliability of disaster communication across different channels are (e.g news, social media post, press). They would be based in Buenos Aires, speakers of Rioplatense Spanish, with an intermediate to advanced level of English, at least a master's degree, and be part of an upper-middle-class household.

Overall Risk Ratings

Benchmark	Risk	Avg. Score
WXIMPACTBENCH	HIGH	2.2 / 5
CRISISBENCH	HIGH	2.3 / 5

Dimension Scores Comparison

Dimension	WXIMPACTBENCH	Rating	CRISISBENCH	Rating	Δ
Input Ontology (IO)	3/5	Moderate gaps	3/5	Moderate gaps	0
Input Content (IC)	1/5	Serious concern	1/5	Serious concern	0
Input Form (IF)	2/5	Significant gaps	3/5	Moderate gaps	-1
Output Ontology (OO)	2/5	Significant gaps	2/5	Significant gaps	0
Output Content (OC)	2/5	Significant gaps	2/5	Significant gaps	0
Output Form (OF)	3/5	Moderate gaps	3/5	Moderate gaps	0
Average	**2.2**		**2.3**		**-0.2**

Δ = WXIMPACTBENCH score − CRISISBENCH score. Positive = regional benchmark scores higher.

Overall Summaries

WXIMPACTBENCH (risk: HIGH)

WXImpactBench is an English-language, formally-styled, North America-centric benchmark whose six top-level impact categories transfer reasonably to Argentine disaster types but whose content, register, language, annotator pool, and output schema all diverge substantially from the deployment's needs. The most severe validity violations are in input_content (zero Spanish-language, zero Argentine, zero social media content; HIGH priority) and output_ontology (no native credibility scoring — though deliberately handled downstream by user design — and no sub-labels for Argentine-specific granularity; HIGH priority). Input_form is mismatched on register and length, and the paper itself acknowledges formal historical text is easier for LLMs than informal modern text, meaning benchmark scores likely overstate deployment-relevant capability. Output_content carries zero Argentine annotator representation and exhibits empirical labeling inconsistencies on positively-framed chunks. The benchmark may serve as one input to model triage but cannot stand alone as evidence of deployment-readiness for Argentine government disaster reporting reliability assessment.

CRISISBENCH (risk: HIGH)

CrisisBench provides a methodologically rigorous, reproducible foundation for crisis-tweet classification with a clean two-task ontology (binary informativeness + multi-class humanitarian type) that aligns at the top level with international ISCRAM/OCHA-derived frameworks Argentine emergency professionals reference. However, three structural misalignments substantially limit its validity for the Buenos Aires policy-maker deployment: (1) Input Content (HIGH priority) is severely misaligned — 94.46% English [Q46], zero Argentine or Rioplatense Spanish content, and a 2010–2017 temporal window that predates the dissolution of Télam (July 2024) and creation of AFE (2025); the only Spanish content in the benchmark is Chilean and Venezuelan, not Rioplatense [DATASET-D2, D3, D6, D7]. (2) Output Ontology (HIGH priority) lacks the source credibility / signal-reliability dimension that is the deployment's core requirement; the user acknowledges this as a downstream construction step but the benchmark provides no learnable credibility signal. (3) Output Content shows annotator-population mismatch and empirically observable annotation noise, with evidence of systematic Spanish-language disaster-content under-recognition [DATASET-D9, D8]. Input Form, Input Ontology, and Output Form present moderate but tractable gaps. The benchmark is best treated as a usable signal-triage foundation for the user's planned downstream credibility-aggregation pipeline, not as a complete evaluation framework for the Argentine deployment.

Per-Dimension Justification Comparison

Each benchmark's full assessment justification and cited evidence, shown side by side.

Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

WXIMPACTBENCH — 3/5 (Moderate gaps) [high confidence]	CRISISBENCH — 3/5 (Moderate gaps) [high confidence]
The six top-level impact categories (Infrastructural, Political, Financial, Ecological, Agricultural, Human Health) are broadly applicable to Argentine disaster types and the elicitation confirms they 'could accommodate the named disaster types' [A1]. Empirical inspection confirms each category appears in the data with positive examples [DATASET-D14, D16, D18]. However, the ontology is constructed entirely around formal newspaper articles [Q18, Q115], with no task type targeting social media, press releases, or short-form informal content — the deployment's highest-priority input channels. No sub-labels exist within the six categories, a recognized gap for Argentine policy granularity (e.g., villa miseria vs. formal infrastructure). The chunked dataset also includes a high proportion (≈65%) of off-topic zero-label fragments [DATASET-D1, D2, D7, D23] that may dilute the discriminative validity of the ontology for genuine disaster classification. The QA task is closed-retrieval only [Q89], precluding the open-retrieval setting most relevant to a deployment scanning real source streams.	The top-level taxonomy (informativeness binary; humanitarian multi-class with categories like affected_individual, infrastructure_and_utilities_damage, requests_or_needs, response_efforts, caution_and_advice) is broadly compatible with the international ISCRAM/OCHA-derived frameworks Argentine emergency professionals use as reference points, and elicitation Q1 confirms users find these categories 'broadly sufficient'. However, the benchmark explicitly defers event-type classification [Q22, Q23], does not include sub-labels for Argentine hazard types (pampero, sudestada, ENSO drought, AMBA urban flash flood) or informal-settlement (villa) impact framing, and the absence of a Latin American taxonomy adaptation is confirmed at the field level [WEB-15]. Some humanitarian classes have very low support (e.g., 17 'terrorism related' tweets [Q52]), and categorical granularity varies significantly across constituent datasets [Q76]. Dataset analysis confirms the schema covers the core triage categories Argentine policy makers need [DATASET-D19, D20, D35, D42] but exposes the 'other_relevant_information' class as a heterogeneous catch-all that limits its utility for borderline credibility cases.

WXIMPACTBENCH evidence	CRISISBENCH evidence
[Q18] 'Climate impact analysis (Thulke et al., 2024) aims to help society make correct decisions about climate-related challenges affecting communities' (p.2)	[Q22] 'We only focused on two tasks for this study and we aim to address event types task in a future study.' (p.2)
[Q89] 'The practical open-retrieval setting, i.e., identifying the relevant articles from a huge database, is left for future studies' (p.9)	[Q52] 'Only 17 tweets with the label "terrrorism related" are present in CrisisNLP.' (p.5)

DATASET-D14: 1894 blizzard chunk correctly labeled infrastructural=1, confirming category is empirically distinguishable	[Q76] 'CrisisLex and CrisisNLP ... six and eleven class labels, respectively' (p.7)
DATASET-D16: 1887 Montreal snow removal chunk labeled infra=1+political=1+financial=1, demonstrating multi-label realism	[WEB-15] TREC-IS 25-category ontology was developed with North American/European emergency practitioners and does not include Argentine hazard types
DATASET-D2, D7, D23: crossword puzzles, film reviews, OCR-corrupted stock tables labeled all-zero — high proportion of off-topic chunks dilute ontology signal	DATASET-D19: Calgary flood tweet correctly labelled affected_individual — confirms top-level category functions
	DATASET-D42: Tacloban evacuation centers tweet labelled response_efforts — clear category alignment

Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

WXIMPACTBENCH — 1/5 (Serious concern) [high confidence]	CRISISBENCH — 1/5 (Serious concern) [high confidence]
The benchmark corpus is exclusively English-language historical Canadian newspaper text from La Presse, La Patrie, and the Montreal Gazette [Q97]. No Spanish-language, Argentine, or Latin American source material is represented, and no Argentine institutional actors (SINAGIR, SMN, AFE), geographic references, or local disaster framing appear. The deployment explicitly handles Rioplatense Spanish across formal press, social media, and informal community channels — none of which are in the benchmark. The paper itself acknowledges 'potential biases in underrepresented historical events and linguistic variations' [Q90] and that 'the interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors' [Q96]. Empirically, the dataset contains some non-Canadian wire-service content (China, Mediterranean, Indonesia) [DATASET-D18, D19, D21], which marginally broadens geographic scope but does not include South America or Spanish-language content. Web search confirmed no Argentine Spanish-language disaster NLP benchmark exists [WEB-14, WEB-15] to supplement this gap. Given that input_content is a HIGH-priority dimension per elicitation, this represents a major validity violation.	Input content is the highest-priority dimension (HIGH per elicitation) and exhibits the most severe alignment failure. English tweets constitute 94.46% of the informativeness set [Q46], with non-English content largely confined to CrisisLex [Q47], and all classification experiments restricted to English [Q59]. The geographic distribution is concentrated in North American and globally reported events: Joplin tornado, Hurricane Sandy [Q27, Q28], 2010/2012 events [Q29], 2017 international disasters [Q31] — with no Argentine or Latin American disaster events documented. Dataset analysis empirically confirms that across 271 sampled examples, zero contain Argentine events, Argentine institutional actors, or Rioplatense Spanish; the only Spanish-language content is Chilean and Venezuelan [DATASET-D2, D3, D4, D6, D7]. The deployment's primary channel — Argentine Twitter/X with Rioplatense voseo, lunfardo, political satire, and references to local actors (SINAGIR, AFE, Buenos Aires province) — is entirely absent from the training distribution. The 2010–2017 temporal window [Q100] predates the dissolution of Télam (July 2024) [WEB-9], creation of AFE (2025) [WEB-10], and post-Musk Twitter/X conventions, meaning the institutional credibility landscape encoded in the data does not match the current Argentine deployment context.

WXIMPACTBENCH evidence	CRISISBENCH evidence
[Q97] 'Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec.' (p.9)	[Q46] 'English tweets appear to be highest in the distribution ... 94.46% of 156,899' (p.5)
[Q90] 'it may have potential biases in underrepresented historical events and linguistic variations' (p.9)	[Q47] 'most of the non-English tweets appear in the CrisisLex dataset' (p.5)
[Q96] 'The interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors' (p.9)	[Q27] 'Joplin collection ... Sandy collection' (p.3)

[WEB-14] HumAID English-language disaster tweets exists but contains no Argentine events or Spanish content	[Q31] 'CrisisMMD ... seven disaster events that happened in 2017' (p.3)
DATASET-D18: Chinese flood content labeled — confirms ontology applied to non-Canadian geography	[Q59] 'we only use tweets in English language in our experiments' (p.6)
DATASET-D19: Mediterranean thunderstorm flooding labeled political=1+ecological=1 — European disaster	[Q100] 'time-span starting from 2010 to 2017' (p.9)
DATASET-D21: Indonesian volcanic eruption labeled — but no South American content observed in 31-example sample	[WEB-9] Télam dissolved by Decree 548/2024 in July 2024 — material change to Argentine formal press source landscape post-benchmark
	[WEB-10] AFE created by Decree 225/2025 — new federal emergency coordinator absent from benchmark institutional context
	DATASET-D2: Chilean earthquake solidarity tweet — Chilean Spanish, not Argentine/Rioplatense
	DATASET-D3: Iquique (Chile) solidarity tweet — confirms Spanish content is exclusively Chilean in sample
	DATASET-D4: Chilean political sentiment about foreign aid labelled not_informative — non-Argentine register
	DATASET-D6: Chilean earthquake tweet — Chilean, not Rioplatense
	DATASET-D7: Venezuelan refinery explosion tweet — Venezuelan Spanish

Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

WXIMPACTBENCH — 2/5 (Significant gaps) [high confidence]	CRISISBENCH — 3/5 (Moderate gaps) [high confidence]
The benchmark is text-only [Q19], which matches the deployment's text-based channels. However, the form characteristics diverge substantially: benchmark inputs are long-form (avg 2,987 tokens) or chunked (avg 781 tokens) [Q109], formally structured historical journalism with 'outdated terminology, spelling variations, and evolving writing conventions' [Q14]. The Argentine deployment includes social media posts (often <280 chars), WhatsApp messages, hashtags, emoji, lunfardo slang, and informal abbreviation — none of which is represented. Critically, the paper itself notes that formal historical narrative style is *easier* for LLMs to classify [Q83, Q85], meaning benchmark performance estimates likely overstate model capability on noisier informal inputs. Web research confirms most LLMs face specific challenges with flood-related social media even in English [WEB-17]. The benchmark is English-only with Latin script; Rioplatense Spanish, while also Latin-script, introduces voseo, yeísmo rehilado lexicon, and Italian-influenced vocabulary not exercised. Empirically, the dataset contains heavily OCR-corrupted financial table chunks [DATASET-D23] which are not analogous to Argentine social media noise but do confirm noise tolerance is partially exercised. Input form is a MODERATE-priority dimension per elicitation; score reflects significant register and length mismatch.	Input form is rated LOWER priority by elicitation, and the structural alignment is moderate. The benchmark is text-only Twitter content [Q38] matching the deployment's text-only modality, and Argentine professional cohorts have high digital infrastructure access [WEB-3, WEB-5]. Tokenization, URL/punctuation/user-id removal [Q38], and near-duplicate filtering at 0.75 cosine similarity [Q39, Q40] are reasonable preprocessing choices. However, three concerns remain: (a) the deployment spans Twitter/X, news articles, and press releases, but the benchmark is exclusively tweet-derived with no cross-channel transfer evaluation [WEB-20]; (b) language tagging is done via Google Cloud API [Q44, Q45] but lang_conf is missing for many CrisisMMD/AIDR examples (Concern 9), weakening language-stratified routing for the bilingual deployment; (c) all classification experiments are English-only [Q59] despite multilingual content being present in the consolidated data [Q42, Q43], so the form-level multilingual capability is not validated. The 70/10/20 split [Q60] and aggressive duplicate filtering [Q41, Q97] are methodological strengths.

WXIMPACTBENCH evidence	CRISISBENCH evidence
[Q19] 'The construction of the dataset aims to obtain high-quality text samples... data collection, post-OCR correction, topic-aware article selection, and manual label annotation.' (p.3)	[Q38] 'modified version of the Tweet NLP tokenizer ... lowercasing the text and removing URL, punctuation, and user id' (p.4)
[Q109] 'The average number of tokens per article is 2987.4 in long-context settings and 781.3 in mixed-context settings.' (p.15)	[Q39] 'cosine similarity to compute a similarity score between two tweets and flag them as duplicate if their similarity score is greater than the threshold value of 0.75' (p.4)
[Q14] 'Historical newspapers often employed more descriptive and elaborate narratives compared to modern reporting styles' (p.2)	[Q42] 'Some of the existing datasets contain tweets in various languages (i.e., Spanish and Italian) without explicit assignment of a language tag' (p.5)
[Q83] 'The reason might be the structured and formal narrative style used to report disruptive weather events in historical periods, which more explicitly highlights cause-and-effect relationships.' (p.8)	[Q44] 'We decided to provide a language tag for each tweet' (p.5)
[Q85] 'the language patterns used in historical narrative styles are easier for language models to identify, and thus perform better on classifying disruptive weather impacts.' (p.8)	[Q45] 'language detection API of Google Cloud Services' (p.5)
[WEB-17] Most LLMs face specific challenges processing flood-related social media data even in English (HumAID-based six-LLM analysis)	[Q59] 'we only use tweets in English language in our experiments' (p.6)

DATASET-D23: OCR-corrupted financial tables present as test rows — exercises noise tolerance but not analogous to social media noise	[Q60] 'train, dev, and test sets with a proportion of 70%, 10%, and 20%' (p.6)
	[WEB-3] 88.4% Argentine national internet penetration as of January 2024
	[WEB-5] AMBA professional cohort fully connected; 71% of fixed-line subscriptions in CABA/Buenos Aires province/Córdoba/Santa Fe/Mendoza
	[WEB-20] TREC CrisisFACTS extends to multi-stream (Twitter, Reddit, Facebook, online news) but provides no tweet-to-news transfer benchmarks

Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

WXIMPACTBENCH — 2/5 (Significant gaps) [high confidence]	CRISISBENCH — 2/5 (Significant gaps) [high confidence]
The deployment's primary decision-support output — reliability/credibility scoring of disaster reports — is structurally absent from the benchmark output space, which produces only six binary impact labels [Q25, Q33, Q117] and ranking lists [Q36]. The user envisions credibility as a downstream layer aggregating model outputs with external metadata [A3], so this gap is partially mitigated by deployment design. However, even within the impact-labeling output, the binary schema with no sub-labels limits granularity for Argentine policy needs (villa miseria vs. formal infrastructure; soybean/maize vs. Canadian grain framing). Annotation guidelines emphasize 'direct and immediate effects' and 'explicit references' [Q118, Q147], excluding inferred or slow-onset impacts — a meaningful constraint for ENSO-driven Pampa drought assessment. Empirically, dataset analysis identifies labeling inconsistencies on positive-content chunks (1998 Quebec ice storm with 1M+ without power labeled all-zero [DATASET-D17]; 1881 storm with explicit casualties labeled human_health=0 [DATASET-D27]) that suggest the binary output decisions may not be fully reliable. Output ontology is HIGH-priority per elicitation; score reflects the structural absence of the credibility dimension and granularity constraints, partially offset by user's deliberate design to handle credibility downstream.	Output ontology is HIGH priority per elicitation. The two implemented tasks — binary informativeness and multi-class humanitarian type [Q12] — provide a usable foundation but do not natively include the source credibility, misinformation likelihood, or signal-to-noise reliability dimension that is the deployment's core requirement (elicitation Q3). The user explicitly acknowledges this gap and intends to construct credibility scoring downstream by combining model outputs with external metadata, but the benchmark's data schema (id, event, source, text, lang, lang_conf, class_label) contains no tweet-level credibility metadata to learn from (CRITICAL Concern 2). Dataset analysis empirically confirms the disconnect: a 'DEATH TOLL Now under Investigation for COVER UP!!!' tweet labelled not_informative [DATASET-D45] and a Boston Bombing conspiracy tweet labelled informative [DATASET-D47] illustrate that informativeness ≠ credibility. The label space is also imbalanced even after consolidation (37.40% 'not humanitarian' [Q56]), with low-support classes [Q52, Q53, Q54] and class-mapping discrepancies that produce results reported only for classes the model can classify [Q88]. TREC-IS provides the closest precedent (Low/Medium/High/Critical priority labels) but covers no Argentine events [WEB-15].

WXIMPACTBENCH evidence	CRISISBENCH evidence
[Q25] 'Six vulnerability-related disruptive weather impacts are defined as the labeling categories, including Infrastructural, Political, Financial, Ecological, Agricultural, and Human Health.' (p.4)	[Q12] 'Informativeness (binary) and Humanitarian type (multi-class) classification' (p.2)
[Q33] 'six ground-truth impacts Yt = {y^i_t}^6_i=1 with binary labels y^i_t ∈ {0, 1}' (p.5)	[Q52] 'Only 17 tweets with the label "terrrorism related" are present in CrisisNLP' (p.5)

[Q117] 'The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts.' (p.16)	[Q56] 'distribution of "not humanitarian" is relatively higher (37.40%) than other class labels' (p.5)
[Q118] 'The guidelines emphasize focusing on direct and immediate effects, ensuring that classifications are based solely on explicit references within the text.' (p.16)	[Q88] 'we report the results of those classes only for which the model is able to classify' (p.8)
[Q147] 'Return false for any impact category that is either not present in the text or not related to weather events. Base classifications on explicit mentions in the text. Focus on direct impacts rather than implications.' (p.20)	[WEB-15] TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events
DATASET-D17: 1998 Quebec ice storm with >1M without power labeled all-zero — potential under-labeling concern for human_health and infrastructural recall	DATASET-D45: 'DEATH TOLL ... COVER UP!!!' tweet labelled not_informative — illustrates informativeness ≠ credibility
DATASET-D27: 1881 storm with skull fracture and child death labeled human_health=0 — labeling inconsistency	DATASET-D47: Boston Bombing conspiracy tweet labelled informative — credibility-disjoint output space

Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

WXIMPACTBENCH — 2/5 (Significant gaps) [high confidence]	CRISISBENCH — 2/5 (Significant gaps) [high confidence]
Annotators are three domain experts from a research group 'specializing in uncovering the history of a region's climate change through the regional historical weather records' [Q110, Q111] — i.e., specialists in Canadian historical climate. No demographic information about annotator nationality, language background, or familiarity with Latin American disaster communication is documented. The paper acknowledges 'the interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors' [Q96] but does not mitigate this. The user does not anticipate systematic disagreement on formal press given Argentine reliance on Global North standards [A4], which moderates concern. However, dataset analysis surfaces concrete labeling inconsistencies: 1998 Quebec ice storm (1M+ without power, -30°C) labeled all-zero on infrastructural and human_health [DATASET-D17]; 1881 storm with explicit fatalities labeled human_health=0 [DATASET-D27]; China flood + Japan deaths labeled infra=1 but human_health=0 [DATASET-D22] while a smaller-scale Indonesian eruption is labeled human_health=1 [DATASET-D21]. These inconsistencies raise convergent validity concerns even within the Canadian context, compounded by the absence of any Argentine validation. Output content is MODERATE-priority per elicitation; score reflects the combination of zero Argentine annotator representation and empirical labeling inconsistencies.	Annotation provenance is minimally documented: the only directly attested annotation protocol is that AIDR data was labeled by domain experts [Q111]. For the remaining seven source datasets, no annotator demographics, recruitment methods, or inter-annotator agreement statistics are reported in the paper. Web research confirms CrisisNLP used crowdsourced workers with three-annotator agreement [WEB-18], but no Argentine annotators or Argentine-norm calibration is documented anywhere in the provenance chain. Cross-dataset evaluation reveals a 14.3% F1 drop when training on CrisisLex and evaluating on CrisisNLP [Q87], demonstrating that label conventions are not fully portable across event contexts — which raises substantial concern about portability to Argentine discourse. Dataset analysis empirically reveals concrete annotation noise: a Chilean cinema tweet labelled displaced_and_evacuations [DATASET-D1], NGO mission descriptions labelled requests_or_needs [DATASET-D37, D38], substantive typhoon warnings labelled not_informative [DATASET-D28, D30], philosophical text labelled informative [DATASET-D11], and Spanish-language disaster warnings systematically labelled not_informative [DATASET-D9, D8] — suggesting potential systematic under-recognition of Spanish-language disaster signals. Elicitation Q4 indicates Argentine professionals largely accept Global North label conventions, which softens but does not eliminate the concern.

WXIMPACTBENCH evidence	CRISISBENCH evidence

[Q110] 'The annotation process was conducted by members of a research group specializing in uncovering the history of a region's climate change through the regional historical weather records.' (p.15)	[Q87] '14.3% difference in F1 on CrisisNLP data using the CrisisLex model for the informativeness task' (p.8)
[Q111] 'Their expertise can ensure the accuracy and reliability of annotations.' (p.15)	[Q111] 'AIDR data has been labeled by domain experts using AIDR system' (p.11)
[Q96] 'The interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors' (p.9)	[WEB-18] CrisisNLP used crowdsourced annotation with three-annotator agreement; demographics not published
DATASET-D17: 1998 Quebec ice storm chunk with 1M+ without power, -30°C, labeled all-zero — potential infrastructural and human_health under-labeling	DATASET-D1: Chilean cinema tweet labelled displaced_and_evacuations — apparent annotation error
DATASET-D27: 1881 storm with skull fracture and child death labeled human_health=0	DATASET-D9: Spanish-language Chilean earthquake aid warning labelled not_informative — systematic Spanish under-recognition risk
DATASET-D22: China flood (810,000 homes collapsed) + Japan E. coli deaths labeled infra=1, human_health=0 — inconsistent with D21	DATASET-D8: Spanish meteorological typhoon warning labelled not_humanitarian — same Spanish-content asymmetry
DATASET-D21: Indonesia minor injuries (20 people) labeled human_health=1 — inconsistent threshold vs. D22	DATASET-D28: Quantitative typhoon warning labelled not_informative — boundary inconsistency
	DATASET-D37: Capability description labelled requests_or_needs — mislabel
	DATASET-D38: NGO mission statement labelled requests_or_needs — mislabel
	DATASET-D11: Philosophical/religious text labelled informative — boundary anomaly

Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

WXIMPACTBENCH — 3/5 (Moderate gaps) [medium confidence]	CRISISBENCH — 3/5 (Moderate gaps) [high confidence]
Output form consists of binary label vectors and ranked article lists [Q33, Q36, Q43–Q45, Q50], producing F1, accuracy, row-wise accuracy, Hit@1, nDCG@5, Recall@5, and MRR. These outputs are usable as upstream features for the deployment's downstream credibility aggregation per A3, so the form is functionally compatible. However, the benchmark does not natively produce the credibility/reliability score that constitutes the deployment's primary decision output, and the transformation from binary labels to credibility scores requires non-trivial integration with Argentine source metadata (outlet reach, account verification, institutional affiliation) that is unaddressed. Row-wise accuracy [Q45, Q75] provides a stricter evaluation that more closely approximates aggregation reliability needs. Output form is MODERATE-priority per elicitation; score reflects functional compatibility tempered by the non-trivial transformation gap and the absence of confidence/probability outputs that would be most useful for downstream aggregation.	Output form is MODERATE priority per elicitation. The benchmark produces discrete classification labels evaluated via weighted precision/recall/F1 [Q75], with documented model families (CNN [Q63], fastText [Q64], BERT/DistilBERT/RoBERTa [Q66]) achieving 0.866 F1 informativeness and 0.829 F1 humanitarian on consolidated English test set [Q84]. Hyperparameters are fully documented [Q113–Q121] and ten re-runs are performed for stability [Q72]. The output form is clean, reproducible, and suitable for downstream pipeline construction. However, two structural gaps remain: (a) the deployment envisions an aggregated continuous credibility score, which the benchmark does not produce — it requires downstream construction (user-acknowledged); no published pipeline combining CrisisBench outputs with external metadata exists [WEB-21]; (b) multilingual evaluation shows ~8% F1 drop on humanitarian classification using BERT [Q99], indicating that even where multilingual output is attempted, performance degrades substantially. Pre-trained models are mainly trained on non-Twitter text, an acknowledged limitation [Q65]. The discrete-to-continuous transformation is a structural gap but is bounded and tractable.

WXIMPACTBENCH evidence	CRISISBENCH evidence
[Q33] 'six ground-truth impacts Yt = {y^i_t}^6_i=1 with binary labels y^i_t ∈ {0, 1}' (p.5)	[Q65] 'pre-trained models are mainly trained on non-Twitter text' (p.6)
[Q43] 'For multi-label classification task, we use F1-score, accuracy, and row-wise accuracy as evaluation metrics.' (p.6)	[Q72] 'we perform ten re-runs for each experiment using different seeds' (p.6)
[Q45] 'the row-wise accuracy is a strict metric that requires more accurate output as the model should correctly classify all six impact labels for a given article.' (p.6)	[Q75] 'weighted average precision (P), recall (R), and F1-measure (F1) ... takes into account the class imbalance problem' (p.7)
[Q50] 'For the ranking-based QA task, we deploy the standard metric that emphasizes the accuracy of top positions for evaluation, including Hit@1, nDCG@5, Recall@5, and MRR.' (p.6)	[Q84] '0.866 (F1) for informativeness and 0.829 for humanitarian' (p.8)
	[Q99] '~8% drop using BERT model' (p.9)
	[WEB-21] 2024 NLP credibility survey: aggregating multiple credibility signals into a unified score remains fragmented and underspecified